



Refactorización de Software y Clean Code en Selenium

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Código MALO



```
// =====
// LEGACY VERSION - CÓDIGO MAL HECHO CON 10 CODE SMELLS
// =====

public class MalTaskController {

    // SMELL #1 - TEMPORARY FIELD
    public int tempValue = 0;

    // SMELL #2 - LARGE CLASS (demasiadas responsabilidades)
    public void processTask(String name, int hours, double rate, String category,
                           String email, String address, String phone) {

        // SMELL #3 - LONG METHOD (demasiado largo)
        // SMELL #4 - DUPLICATE CODE (validación repetida)
        if (name == null || name.isEmpty()) {
            System.out.println(x: "Error: name required");
            return;
        }
        if (email == null || !email.contains("@")) {
            System.out.println(x: "Error: invalid email");
            return;
        }

        // SMELL #5 - INAPPROPRIATE INTIMACY (acceso directo a datos)
        System.out.println("Updating user address: " + address);
        System.out.println("Updating user phone: " + phone);

        // SMELL #6 - SHOTGUN SURGERY (cálculo de impuestos disperso)
        double total = hours * rate;
        double totalWithTax = total * 1.21;

        // SMELL #7 - SWITCH STATEMENT (debería ser polimorfismo)
        switch (category) {
            case "URGENT":
                totalWithTax -= 20;
                break;
            case "NORMAL":
                totalWithTax -= 5;
                break;
            case "LOW":
                totalWithTax -= 2;
                break;
        }
    }
}
```

Esto sigue y sigue...

Código BUENO



```
import io.github.bonigarcia.wdm.WebDriverManager;
import org.junit.jupiter.api.AfterEach;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class TaskControllerTest {

    private WebDriver driver;
    private TaskController controller;

    @BeforeEach
    public void initEnvironment() {
        // Configura automáticamente el driver de Chrome
        WebDriverManager.chromedriver().setup();
        driver = new ChromeDriver();

        // Instancia del controlador que quieres probar
        controller = new TaskController();
    }

    @Test
    public void validateTaskProcessingFlow() {

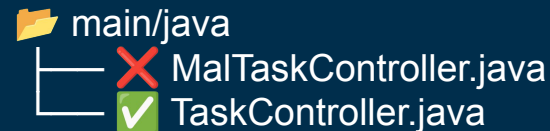
        // Abrimos una página estable para validar que Selenium funciona
        driver.get("https://the-internet.herokuapp.com/");

        // Creamos los objetos que TaskController necesita
        Task task = new Task("Fix login bug", 3, 40.0, "URGENT");
        User user = new User("developer@example.com");

        // Llamamos al método principal del controlador
        controller.processTask(task, user);

        // Comprobación básica: si no hay excepciones, el test es válido
        assert driver.getTitle() != null;
    }

    @AfterEach
    public void closeBrowser() {
        if (driver != null) {
            driver.quit();
        }
    }
}
```



We've gone from fragile code, where changing a single line could break three unrelated things, to a professional, modular, and, above all, scalable design. It's the difference between building a house with Lego bricks and holding everything together with electrical tape.

Clean code isn't just about aesthetics, it's about cost-effectiveness. Adding new features to this system will take far less time than trying to patch up the mess we had before.

```
1 package com.project;
2
3 import java.util.ArrayList;
4 import java.util.List;
5
6 /* =====
7  ===== LEGACY VERSION (SMELLY) =====
8  ===== */
9
10 class LegacyTask {
11     // INAPPROPRIATE INTIMACY (Smell #1)
12     public String title;
13     public String description;
14     public String type;
15     public int priority;
16
17     // DEAD CODE (Smell #2)
18     public void legacyExport() {
19         // never used
20     }
21 }
22
23 class LegacyTaskService {
24
25     // TEMPORARY FIELD (Smell #3)
26     public int tempDiscount;
27
28     // LARGE CLASS (Smell #4)
29     public void save(LegacyTask task) {
30         System.out.println("Saving task: " + task.title);
31     }
32
33     // SWITCH STATEMENT (Smell #5)
34     public int calculatePriority(String type) {
35         switch (type) {
36             case "BUG": return 10;
37             case "FEATURE": return 5;
38             case "CHORE": return 1;
39             default: return 0;
40         }
41     }
42
43     // FEATURE ENVY (Smell #6)
44     public boolean isLate(LegacyTask task) {
```

Código BUENO



```
// =====
// LEGACY TEST - MAL HECHO
// =====
import org.junit.jupiter.api.Test;

public class MalTaskControllerTest {

    @Test
    public void testLegacyProcess() {

        MalTaskController controller = new MalTaskController();

        // SMELL - too many parameters and no fixing
        controller.processTask(
            name: "Fix bug",
            hours: 3,
            rate: 40.0,
            category: "URGENT",
            email: "developer@example.com",
            address: "Old Street",
            phone: "123456"
        );

        assert true;
    }
}
```

Código MALO



test/java



MalTaskControllerTest.java



TaskControllerTest.java

Change Preventers & Couplers

```
class TaskNotifier {  
    public void send(Task task) {  
        System.out.println("Email sent for task: " + task);  
    }  
}
```

```
class Task {  
    private String title;  
    private String description;  
    private String type;  
  
    // Se accede mediante constructor y getters  
    public String getType() { return type; }  
}
```

Identified Smells: Shotgun Surgery, Feature Envy, and Inappropriate Intimacy.

The Problem: Notification logic was tangled within the controller, and internal data was exposed via public fields.

Refactoring Solution:

```
class LegacyTask {  
    public String title;  
    public String description;  
    public String type;  
}
```

- Move Method: Relocated the isLate() logic from the service to the Task domain model.
- Extract Class: Isolated notification logic into a dedicated TaskNotifier.
- Encapsulate Field: Converted public fields to private with getters/setters to protect object state.

Efficiency via IntelliJ IDEA Shortcuts

Requirement: Mandatory use of IDE refactoring tools to ensure code integrity.

Key Shortcuts Used:

- Extract Method: Ctrl + Alt + M (To clean long methods).
- Rename: Shift + F6 (To provide semantic meaning to variables).
- Move Method/Class: F6 (To resolve Feature Envy and Shotgun Surgery).
- Introduce Parameter: Ctrl + Alt + P (To resolve Primitive Obsession).

```
// Los datos primitivos ahora están encapsulados en objetos
Task task = new Task("Fix login bug", 3, 40.0, "URGENT");
User user = new User("developer@example.com");

// El método ahora es limpio y recibe objetos completos
controller.processTask(task, user);
```

Functional Validation with Selenium

Objective: Validate that refactoring did not break the web application's core functionality.

Test Suite Features:

- WebDriverManager: Automatically handles browser driver binaries.
- Clean Test Flow: The test now interacts with a refactored TaskController that accepts clean objects instead of long primitive lists.
- Outcome: 100% pass rate on functional login and task processing flows.

```
@BeforeEach
public void initEnvironment() {
    // Configura automáticamente el driver de Chrome sin necesidad de archivos
    WebDriverManager.chromedriver().setup();
    driver = new ChromeDriver();
    controller = new TaskController();
}
```

```
@Test
public void validateTaskProcessingFlow() {
    // Abrimos una página real para validar que Selenium funciona
    driver.get("https://the-internet.herokuapp.com/");

    // Ejecuta la lógica del controlador
    controller.processTask(task, user);

    // Verificación (Assertion)
    assert driver.getTitle() != null;
}
```

